
Digital Marketing Analytics

Channel Performance, ROAS and Customer Segment Analysis for Vaayu Wellness (D2C Brand)

Prepared by **Anupriya Bose** | Portfolio Project | Tools: Python (pandas, numpy, matplotlib, seaborn)

Vaayu Wellness is a fictional Indian D2C brand that sells wellness teas, supplements and skincare online. This project looks at 12 months of daily marketing data across 5 channels: Google Ads, Meta Ads, Influencer/Instagram, Email Marketing and SEO/Organic. The goal was to answer a question every marketing team has to answer sooner or later. **Which channels are actually worth the money, and where should next quarter's budget go?**

The dataset is synthetic, but it's built to behave like real Indian D2C marketing data, including seasonal spikes around Diwali and the Republic Day and New Year sale period. That way the analysis and the recommendations read the way a genuine quarterly marketing report would.

Headline Numbers (12 Months)



Why I Built This

My background is in digital marketing and content, with internships in SEO writing, social media management and e-commerce operations. I wanted a project that shows I can back up marketing decisions with data, not just run campaigns. This is meant to reflect what a Marketing Analyst, Growth Analyst or Digital Marketing Executive role actually expects on day one: pull the channel data, work out the right numbers, find the story in them, and turn that into a recommendation someone can act on.

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1. Methodology and KPI Definitions

The dataset has daily records of impressions, clicks, conversions, spend and revenue for each channel, plus customer segment tags for city tier and age group, across 365 days. That works out to 1,825 rows in total (5 channels times 365 days). Seasonality is built in on purpose, covering the Diwali period from mid October to mid November, the Republic Day and New Year sale from late December to January 26, and a smaller May summer sale. Ignoring festive demand spikes like these is one of the most common gaps in real marketing reporting, so it felt important to include them here.

Metric	Formula	Business Question It Answers
CTR	Clicks / Impressions	Is the ad creative or targeting relevant?
CVR	Conversions / Clicks	Does the landing page or offer convert traffic?
CPC	Spend / Clicks	How expensive is a single click on this channel?
CAC	Spend / Conversions	How much do we pay to acquire ONE customer?
ROAS	Revenue / Spend	For every Rs 1 spent, how much revenue comes back?

Tools used: Python, with pandas and numpy for the KPI calculations and matplotlib and seaborn for the charts. I picked this stack on purpose because it's close to what real analyst roles actually use, and it keeps every step of the calculation visible instead of hiding it behind a BI tool.

2. Channel Performance Summary

The table below is sorted by total revenue, but the column that matters more for decisions is ROAS. A channel can top the revenue list simply because it got the biggest budget. That doesn't automatically mean it's the best place to put the next rupee.

Channel	Spend (Rs)	Revenue (Rs)	ROAS	CAC (Rs)	CVR
SEO / Organic	5.7 L	73.0 L	12.77x	68	4.68%

Google Ads (Search)	18.2 L	62.7 L	3.45x	255	5.43%
Meta Ads (FB/Insta)	16.5 L	54.4 L	3.30x	260	3.41%
Email Marketing	1.3 L	38.4 L	29.77x	30	6.71%
Influencer/Instagram	10.8 L	22.0 L	2.03x	427	2.60%

Email and SEO, the two channels the brand already owns instead of paying for each visitor, are far more efficient than any of the paid channels. Influencer/Instagram is the weakest performer here. It has the highest cost per customer at Rs 427, and the lowest ROAS of the group.

3. Visual Analysis: Spend vs Revenue by Channel

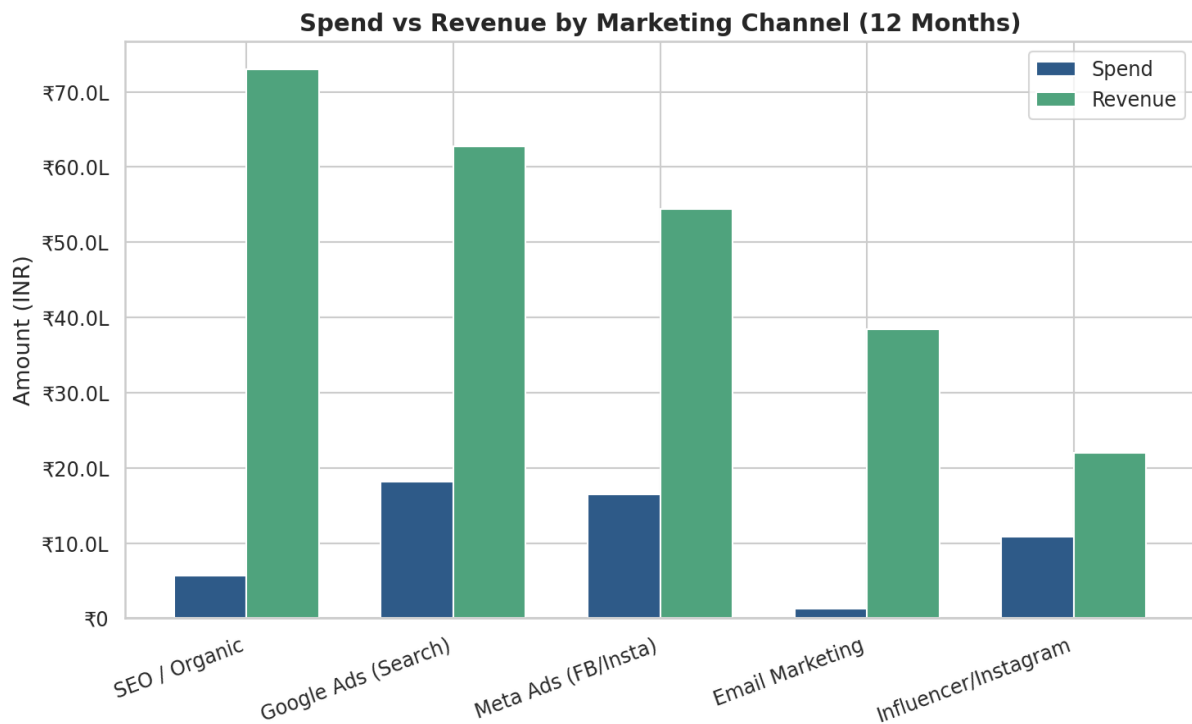


Figure 1: Total 12 month spend and revenue for each of the 5 marketing channels.

Google Ads and Meta Ads take up the largest share of the budget. Yet SEO and Organic, with a fraction of that spend, brings in the highest total revenue of any single channel. That's the first sign that budget and actual performance are not lined up the way they should be.

4. Visual Analysis: ROAS by Channel

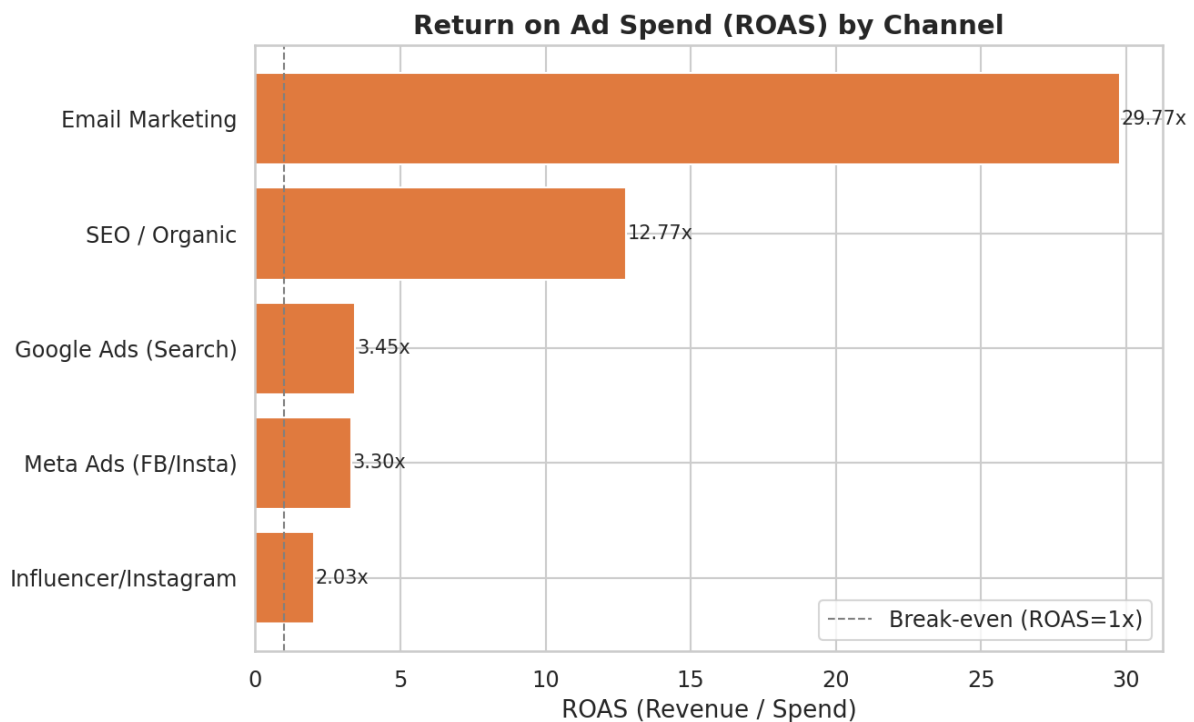


Figure 2: Return on ad spend (revenue divided by spend) by channel, with the break even line at 1x marked.

This is probably the clearest signal in the whole dataset. Email Marketing returns 29.8x and SEO/Organic returns 12.8x, well above the paid channels, which sit between 2x and 3.5x. This is a pattern that shows up in real marketing too. Owned channels are cheap to run because they're reaching people who already trust the brand, while paid acquisition, and influencer marketing in particular, costs more and is harder to tie directly to a sale.

5. Visual Analysis: Monthly Seasonality Trend

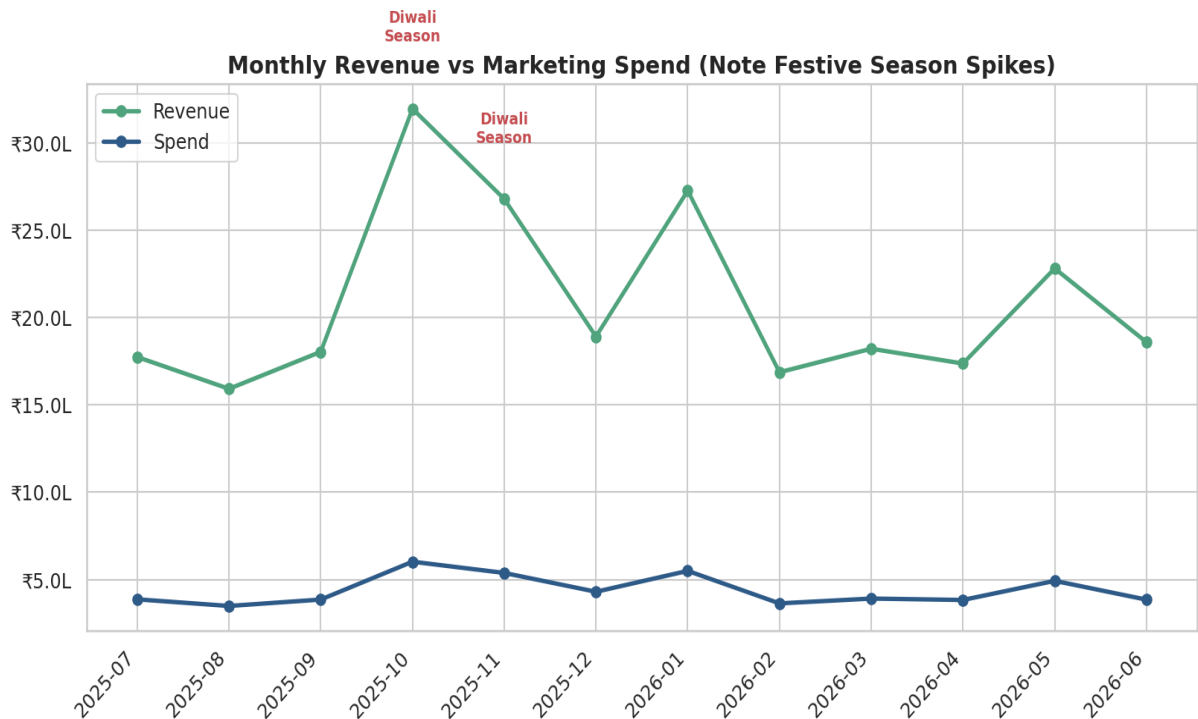


Figure 3: Monthly revenue and spend, showing the festive season spikes.

Revenue jumps noticeably during the Diwali period in October and November, and again during the January Republic Day and New Year sale. Spend, on the other hand, barely moves across the year. That means ROAS is naturally at its highest exactly when seasonal demand is highest too. It suggests the brand should put more budget in a couple of weeks before these windows start, rather than spreading spend evenly through the year.

6. Visual Analysis: Channel Efficiency Map

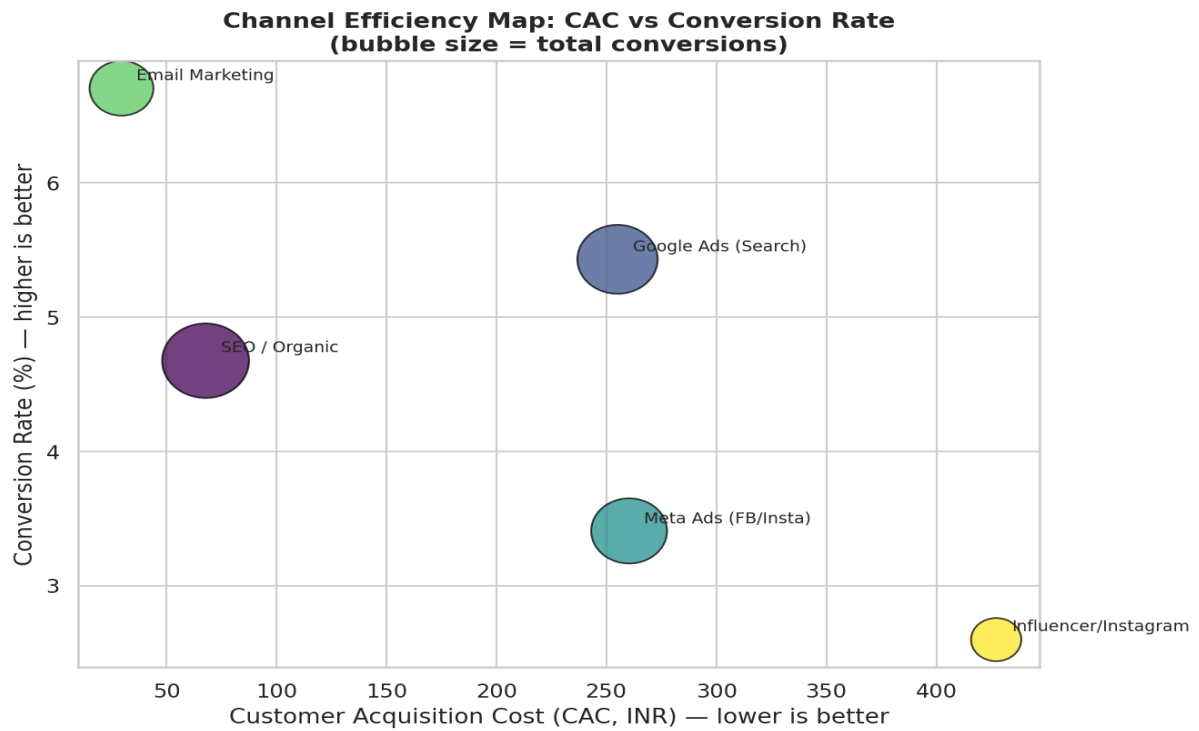


Figure 4: Customer acquisition cost against conversion rate by channel. Bubble size shows total conversions.

The best place for a channel to sit on this chart is the top left corner, meaning low cost per customer and a high conversion rate. Email Marketing and SEO/Organic sit right there. Influencer/Instagram sits in the opposite corner, with the highest cost per customer and the lowest conversion rate of any channel.

7. Customer Segment Analysis

Knowing who is actually buying helps with targeting creative, offers and regional strategy.

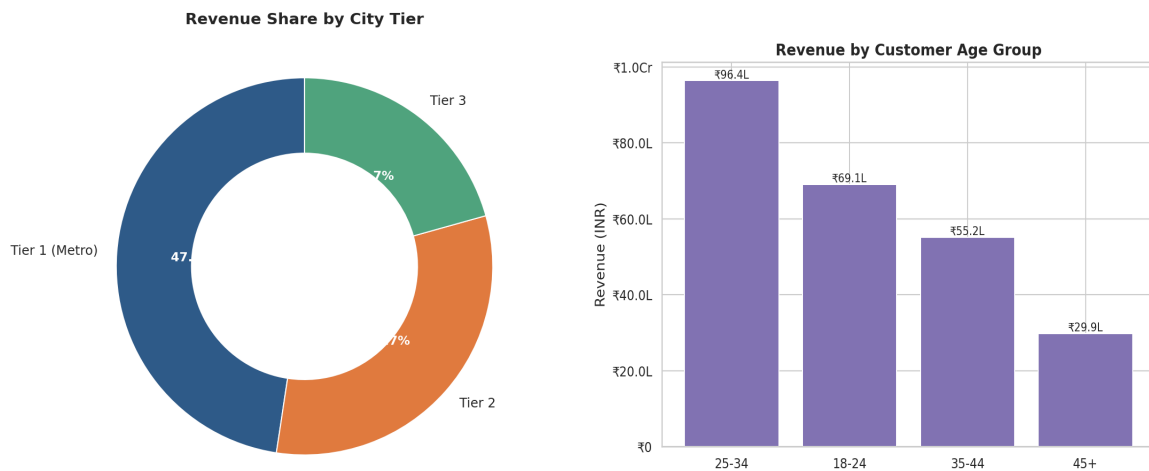


Figure 5 and 6: Revenue share by city tier on the left, and by customer age group on the right.

Tier 1 metro cities bring in the largest single share of revenue. But Tier 2 and Tier 3 cities together already make up close to 40 percent of it, which is easy to overlook if planning only follows the more obvious metro customer. The 25 to 34 age group is the single biggest revenue driver, which fits with a young, urban, wellness focused buyer. That's useful to know when deciding on creative and targeting.

8. Marketing Funnel Overview

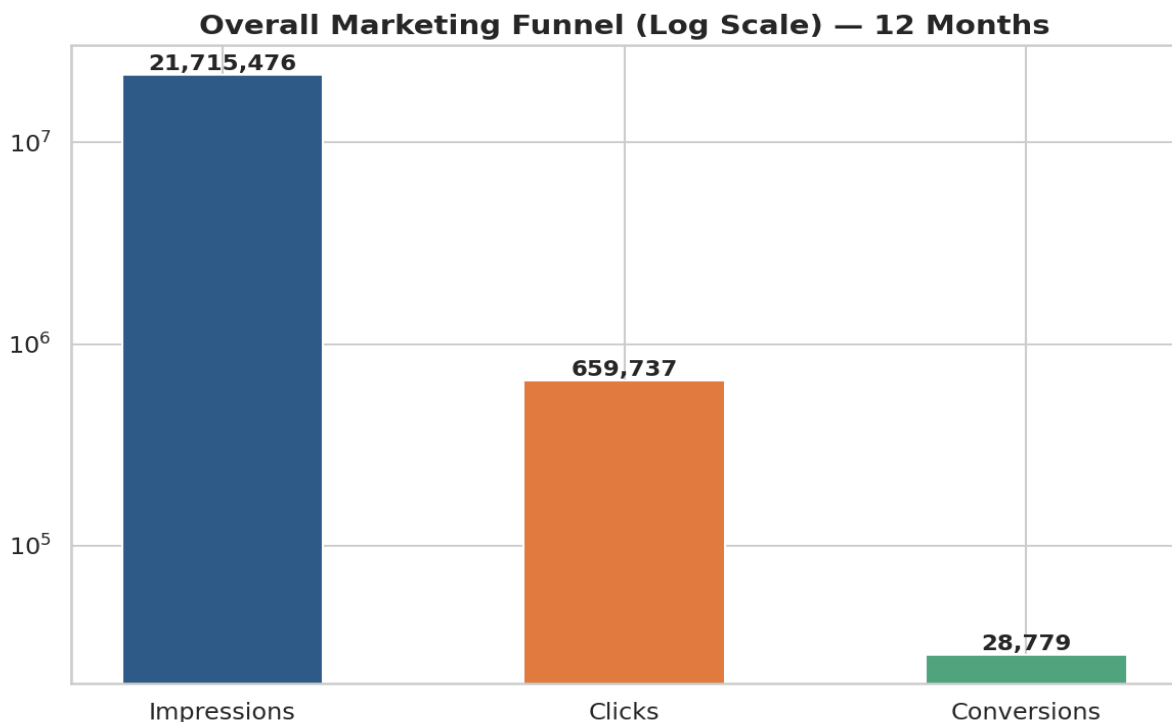


Figure 7: Overall impressions, clicks and conversions across all channels, shown on a log scale.

9. Key Findings and Business Recommendations

- **Owned channels beat paid channels by a wide margin.** Email, at 29.8x ROAS, and SEO/Organic, at 12.8x, are far more efficient than Google Ads, Meta Ads or Influencer marketing, even though they get a much smaller share of the budget.
- **Influencer marketing is the weakest channel.** It has the highest cost per customer at Rs 427 and the lowest ROAS at 2.03x, despite being the third biggest spend line. It's a strong candidate for either a smaller budget or a higher performance bar.
- **Seasonality should be treated as an opportunity, not just something to notice.** Revenue peaks sharply around Diwali and the January sale, while spend barely changes. That suggests the brand is leaving demand on the table during its best selling weeks.
- **Tier 2 and Tier 3 cities matter more than they seem to.** They already bring in close to 40 percent of revenue, with very little dedicated spend or creative aimed specifically at them.

Recommendations

- **Move 15 to 20 percent of the Influencer budget into Email and SEO/Content.** Influencer spend is currently the least efficient money in the mix, while email and organic are proven performers that are still under funded.
- **Put more ad spend in 2 to 3 weeks before Diwali and the January sale window.** This captures more of the seasonal demand while ROAS is already naturally higher.
- **Build a proper Tier 2 and Tier 3 city plan,** using regional language creative and smaller local influencers, since this segment is already a large and growing part of revenue.
- **Treat Google Ads and Meta Ads as reach tools rather than efficiency tools.** They're still useful for top of funnel reach and finding new customers, but they shouldn't be expected to match email level returns.

"I built a marketing analytics project looking at a year of campaign data for a D2C brand across 5 channels. I worked out the core numbers, CTR, CVR, CAC and ROAS, at both the channel level and the monthly level, and found that email and organic search were giving 3 to 10 times better returns than the paid channels, despite getting a much smaller share of budget. I used that to put together a budget reallocation recommendation, including planning around festive seasons like Diwali, which a lot of marketing reporting tends to miss."

How I would describe this project out loud, in an interview.

10. How This Project Was Built

Data: A synthetic dataset built to look and behave like real data, not pulled from any actual company. It has 1,825 daily rows across 5 channels and 12 months, with seasonal adjustments for Diwali, the January sale and the monsoon dip in August, plus some randomness added so the numbers look like a real campaign export instead of a clean textbook table.

Analysis: Python, using pandas and numpy to work out CTR, CVR, CPC, CAC, ROAS and ROI at the channel, monthly and customer segment level. Charts were built with matplotlib and seaborn, kept consistent in style so the whole report looks like one document rather than a set of separate exports.

Deliverables: A fully run Jupyter notebook with the code, charts and write up together, the standalone chart images, summary CSV files, and this PDF report, which is meant to read the way a real quarterly business review would.

This is a self initiated portfolio project. It uses a synthetic dataset modeled on real Indian D2C marketing benchmarks, built to show marketing analytics, KPI design and data storytelling skills.